



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Name :

PRN : 250701220

Class : CSE

Batch : 2025-29

Assignment :

Subject: Programming with Java

ASSIGNMENT No : 5

TITLE :Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.



Code and Output Screenshots

```
J ArrayListExample.java > Language Support for Java(TM) by Red Hat > ArrayListExample
1  import java.util.ArrayList;
2  public class ArrayListExample {
    Run main | Debug main | Run | Debug
3      public static void main(String[] args) {
4          ArrayList<String> list = new ArrayList<>();
5          list.add("Apple");
6          list.add("Banana");
7          list.add("Cherry");
8
9          System.out.println("Fruits: " + list);
10         list.remove("Banana");
11         System.out.println("After removing Banana: " + list);
12
13         System.out.println("Size of the list: " + list.size());
14         System.out.println("Contains Apple? " + list.contains("Apple"));
15     }
16 }
17 }
18 }
```

PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeText, t...) Code

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac ArrayListExample.java && java ArrayListExample
Fruits: [Apple, Banana, Cherry]
After removing Banana: [Apple, Cherry]
Size of the list: 2
Contains Apple? true
```

```
J VectorEg.java > ...
1  import java.util.Vector;
2  public class VectorEg {
    Run main | Debug main | Run | Debug
3      public static void main(String[] args) {
4          Vector<String> vector = new Vector<>();
5          vector.add("Java");
6          vector.add("Python");
7          vector.add("C++");
8
9          System.out.println("Vector Elements: " + vector);
10         vector.remove("Python");
11         System.out.println("After removing Python: " + vector);
12
13         System.out.println("Size of the vector: " + vector.size());
14         System.out.println("Contains Java? " + vector.contains("Java"));
15     }
16 }
17 }
```

PROBLEMS 11 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeText, t...) Code

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac VectorEg.java && java VectorEg
Vector Elements: [Java, Python, C++]
After removing Python: [Java, C++]
Size of the vector: 2
Contains Java? true

[Done] exited with code=0 in 0.513 seconds
```



```
J Strbfr.java > Language Support for Java(TM) by Red Hat > Strbfr
1 public class Strbfr {
    Run main | Debug main | Run | Debug
2 public static void main(String[] args) {
3     StringBuffer sb = new StringBuffer("Hello");
4
5     System.out.println("Original StringBuffer: " + sb);
6
7     sb.append(" World");
8     System.out.println("After append: " + sb);
9
10    sb.insert(5, " Java");
11    System.out.println("After insert: " + sb);
12
13    sb.delete(5, 10);
14    System.out.println("After delete: " + sb);
15
16    sb.reverse();
17    System.out.println("After reverse: " + sb);
18 }
19 }
```

PROBLEMS 11 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !excludeText, t...) Code

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Strbfr.java && java Strbfr
Original StringBuffer: Hello
After append: Hello World
After insert: Hello Java World
After delete: Hello World
After reverse: dlroW olleH
```

Conclusion

ArrayList, Vector, and StringBuffer are useful Java classes for managing dynamic data and strings efficiently. ArrayList provides fast and flexible data storage, Vector offers thread safety, and StringBuffer enables efficient string modification. Choosing the appropriate class improves the performance and reliability of Java applications.

Exercise

1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.



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J TodoList.java > Language Support for Java(TM) by Red Hat > TodoList

```
1  import java.util.ArrayList;
2
3  public class TodoList {
4      |
5      | Run main | Debug main | Run | Debug
6      | public static void main(String[] args) {
7          |
8          | // ArrayList to store tasks
9          | ArrayList<String> tasks = new ArrayList<>();
10         |
11         | tasks.add("Study");
12         | tasks.add("Complete Assignments");
13         | tasks.add("Go to Gym");
14         | tasks.add("Have dinner");
15         | tasks.add("Sleep");
16         |
17         | // StringBuffer to display tasks
18         | StringBuffer sb = new StringBuffer();
19         |
20         | for (String task : tasks) {
21         | | sb.append(task + "\n");
22         | | }
23         |
24         | System.out.println(sb);
25         | }
26     }
```

PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\LENOVO\Documents\JAVA Assignments> javac TodoList.java
● PS C:\Users\LENOVO\Documents\JAVA Assignments> java TodoList
Study
Complete Assignments
Go to Gym
Have dinner
Sleep
❖ PS C:\Users\LENOVO\Documents\JAVA Assignments> |
```



2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```
1 import java.util.ArrayList;
2 public class Q2 {
3     Run main | Debug main | Run | Debug
4     public static void main(String[] args) {
5
6         ArrayList<String> courses = new ArrayList<>();
7
8         // Add courses
9         courses.add("Java");
10        courses.add("Data Structures");
11        courses.add("DBMS");
12        courses.add("Discrete Maths");
13        courses.add("Web Development");
14        courses.add("Operating Systems");
15        courses.add("Python");
16
17        // Remove a course
18        courses.remove("Python");
19
20        // Display registered courses
21        StringBuffer sb = new StringBuffer();
22        sb.append("Registered Courses:\n");
23
24        for (String course : courses) {
25            sb.append("- ").append(course).append("\n");
26        }
27
28        System.out.println(sb);
29    }
```

PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, excludeText, t...

Code

[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Q2.java && java Q2

Registered Courses:

- Java
- Data Structures
- DBMS
- Discrete Maths
- Web Development
- Operating Systems